

POLYMARKET

Trading Bot Landscape

Strategy Research & Architecture Analysis

March 2026

1. Infrastructure Foundation

What every Polymarket bot is built on

All trading bots interact with Polymarket through the same core infrastructure: a Central Limit Order Book (CLOB) running on Polygon blockchain. Understanding this layer is essential before evaluating any strategy.

Core Technical Specs

- Rate limit: 60 orders per minute per API key
- Settlement: Hybrid -- off-chain order matching, on-chain settlement (Polygon)
- Official Python SDK: `py-clob-client`
- Official TypeScript SDK: `@polymarket/clob-client`
- WebSocket streams: real-time orderbook data (6 parallel connections supported)
- Gamma API: separate endpoint for market metadata and event data
- Two market types: binary YES/NO, multi-outcome

Market Mechanics

Each market resolves to either \$1.00 (YES wins) or \$0.00 (NO wins). Shares are priced between 0 and 1, representing the implied probability. For example, YES at 0.62 means the market prices a 62% chance of YES. A YES share + NO share always = \$1.00 at resolution, creating the foundation for arbitrage strategies.

Polymarket offers maker rewards -- bots that provide liquidity by posting limit orders earn a fee rebate on top of the spread they capture. This makes market making doubly attractive for well-positioned bots.

2. Strategy Breakdown

Six distinct approaches found across open-source repositories

1. Market Making

Repos	warproxxx/poly-maker, lorine93s/polymarket-market-maker-bot, terrytrl100/polymarket-automated-mm
Edge Source	Spread capture + Polymarket LP maker rewards
Complexity	High
Risk Level	Medium (inventory imbalance risk)

Description

Market makers post simultaneous limit orders on both YES and NO sides of a contract. They stay inventory-neutral and profit by capturing the bid-ask spread repeatedly. Polymarket's maker reward program adds an additional fee rebate for liquidity providers.

How It Works

1. Fetch current orderbook state via WebSocket.
2. Calculate optimal bid/ask prices based on mid-price and desired spread.
3. Place limit orders on both sides simultaneously.
4. On fill: cancel opposite side, replace both quotes at new prices.
5. Monitor inventory; hedge or pause if one side dominates.
6. Periodically merge YES+NO positions into \$1.00 for redemption (reduces gas).
7. warproxxx/poly-maker fetches all parameters from Google Sheets -- no redeploy needed.

Key Technical Detail

terrytrl100/polymarket-automated-mm calculates order placement using Polymarket's exact maker reward formula, ensuring the spread captured always exceeds gas + reward threshold. It also auto-selects which markets to make based on volume and reward rate.

2. YES+NO Combinatorial Arbitrage

Repos	ent0n29/polybot, Now-Or-Neverr/polymarket-trading-bot
Edge Source	Mispricing: YES + NO combined price < \$1.00
Complexity	Medium
Risk Level	Low (near risk-free if both legs fill)

Description

When the sum of YES price + NO price falls below \$1.00, buying both is guaranteed profit since they resolve to exactly \$1.00 together. In multi-outcome markets, buying all outcomes when their sum < \$1.00 achieves the same guaranteed return.

How It Works

1. Continuously scan all active markets for $\text{price_YES} + \text{price_NO} < 1.00$.
2. Calculate profit: $\text{profit} = 1.00 - (\text{price_YES} + \text{price_NO}) - \text{gas_fees}$.
3. If $\text{profit} > \text{threshold}$: submit simultaneous FOK (Fill or Kill) orders on both sides.
4. FOK ensures both legs fill or neither does -- no directional exposure.
5. Wait for market resolution to collect \$1.00 per pair.
6. Multi-outcome: same logic across all outcome tokens.

Key Technical Detail

The main risk is execution: if only one leg fills (e.g. FOK fails), the position becomes directional. Sophisticated bots submit both orders atomically or cancel both on partial failure. Between April 2024-2025, this strategy class earned an estimated \$40M on Polymarket.

3. Cross-Market Convergence Arbitrage

Repos	dylanpersonguy/Polymarket-Trading-Bot, ent0n29/polybot
Edge Source	Logically correlated markets with diverging prices
Complexity	High
Risk Level	Medium (requires valid logical relationship identification)

Description

Finds pairs or groups of markets where outcomes are logically linked, then exploits price divergences between them. Example: 'Trump wins election' and 'Trump is Republican nominee' are correlated -- winning requires being nominated. If nominee trades at 48c but winner at 52c, there is a 4% arbitrage opportunity.

How It Works

1. Maintain a graph of related markets (political, sports, economic events).
2. Monitor prices across all markets in real-time via WebSocket.
3. Detect divergences exceeding a threshold relative to logical constraints.
4. Enter opposing positions: long the underpriced, short the overpriced.
5. Exit when prices converge or markets resolve.
6. dylanpersonguy's bot scans 1,500+ markets simultaneously with async architecture.

Key Technical Detail

This is the most alpha-rich but hardest-to-automate strategy. Identifying valid logical relationships requires domain knowledge (politics, sports rules, etc.). The most profitable implementations combine automated scanning with human-curated market relationship graphs.

4. CEX-Lag Momentum (Crypto Markets)

Repos	discountry/polymarket-trading-bot, ThinkEnigmatic/polymarket-bot-arena
Edge Source	Polymarket crypto prices lag Binance/Coinbase by seconds
Complexity	Low to Medium
Risk Level	Medium-High (latency arms race)

Description

Targets Polymarket's BTC/ETH/SOL 5-min and 15-min Up/Down binary markets. These markets price whether BTC (or ETH/SOL) will be higher or lower than the current price by expiry. When the underlying asset moves decisively on a CEX, Polymarket prices lag -- this gap is the edge.

How It Works

1. Subscribe to real-time price feeds from Binance, Coinbase, and/or Chainlink.
2. Monitor BTC delta (price change) over the current round window.
3. When momentum exceeds a conviction threshold (e.g. +0.5% in 3 min), enter the UP market.
4. Place limit orders slightly better than current Polymarket price to ensure fill.

5. discountry's bot: detects probability drops >5% in 15min as primary signal.
6. polymarket-bot-arena: 4 bots (momentum, mean-reversion, sentiment, hybrid) compete with Bayesian learning and evolutionary replacement every 12 hours.

Key Technical Detail

Bot Arena Architecture (ThinkEnigmatic): arena.py handles market discovery, trading, resolution, and evolution. learning.py runs a Bayesian engine that tracks win rates by (price bucket, BTC momentum, time of day). Every 12h: bottom 2 bots are killed and replaced with mutated versions of top performers -- continuously self-improving without human intervention.

5. Copy Trading / Whale Tracking

Repos	iengineer/polymarket--whale-trading-bot, dev-protocol/polymarket-copytrading-bot-sport
Edge Source	Mirror profitable whale wallets in real-time
Complexity	Low
Risk Level	High (latency, front-running, whale strategy drift)

Description

Monitors target wallet addresses on Polygon blockchain and automatically replicates their trades with proportional sizing. The thesis: identified whale wallets have a persistent edge that can be copied before their trades fully move the market price.

How It Works

1. Define a list of target whale wallets (identified by historical profitability analysis).
2. Poll wallet positions every ~4 seconds OR listen to on-chain blockchain events.
3. Detect new trades: compare current positions to previous snapshot.
4. Evaluate risk filters: max exposure per market, cooldown, min trade size.
5. Calculate proportional size: $(\text{your_balance} / \text{whale_balance}) \times \text{whale_trade_size}$.
6. Execute via CLOB API immediately.
7. Apply trailing stop-loss tracking peak prices for exit management.

Key Technical Detail

The core challenge is latency: by the time the copy bot detects and executes, the whale's trade may have already moved the market. The best bots use on-chain event listeners rather than polling, reducing detection lag from ~4 seconds to sub-second.

6. AI / LLM-Driven Agent

Repos	Polymarket/agents (official), IISourcell/Poly-Trader, BlackSky-Jose/PolyMarket-trading-AI-model
Edge Source	Information edge via news analysis and LLM probability estimation
Complexity	High
Risk Level	High (model hallucination, prompt sensitivity)

Description

Uses a large language model (GPT-4, Claude, etc.) to estimate the true probability of market outcomes based on current news and context. Compares LLM estimate against market price to find mispricings. The official Polymarket/agents repo is the reference implementation.

How It Works

1. GammaMarketClient fetches active, tradable markets and their metadata.
2. News connector retrieves recent articles relevant to each market question.
3. ChromaDB vectorizes news for semantic retrieval -- LLM gets only most relevant context.
4. LLM is prompted: 'Given this news, what probability resolves YES?'

5. LLM outputs probability + confidence level.
6. If LLM probability > market price by threshold (e.g. LLM=70%, market=55%): place BUY.
7. MongoDB logs all LLM queries, trades, and outcomes for analysis.
8. All actions accessible via cli.py interface.

Key Technical Detail

Architecture is fully modular: connectors (data sources, order types) can be swapped independently. ChromaDB prevents hallucination by grounding the LLM in retrieved facts rather than parametric memory. The agent compares its probability estimate to the market's implied probability and only bets when the gap exceeds a configured threshold.

3. Strategy Comparison

Side-by-side evaluation of all six approaches

Strategy	Edge Source	Complexity	Risk	Reward
Market Making	Spread + LP rewards	High	Medium	Medium
YES+NO Arbitrage	Mispricing	Medium	Low	Low
Cross-Market Arb	Convergence	High	Medium	Medium
CEX-Lag Momentum	Latency vs CEX	Low-Med	Med-High	High
Copy / Whale	Whale alpha	Low	High	Variable
AI / LLM Agent	Info edge	High	High	High

Key Takeaways

- The CLOB API (py-clob-client) is the execution layer for all strategies -- mastering it is step one.
- CEX-lag momentum is the mechanically simplest strategy with a documented, consistent edge in crypto markets.
- YES+NO arbitrage is the lowest-risk strategy but opportunities are thin and quickly competed away.
- Market making earns fees + maker rewards but requires careful inventory management and risk controls.
- Copy trading is easy to implement but latency-sensitive; on-chain event listeners outperform polling.
- LLM agents are architecturally the most sophisticated and hardest to validate -- require rigorous backtesting.
- Cross-market arbitrage has the highest potential alpha but requires domain-specific market relationship knowledge.
- Paper trading mode (polymarket-paper-trader on GitHub) allows risk-free strategy testing before live deployment.

Market Volume Context (2025-2026)

Polymarket processed \$3.74 billion in monthly volume in November 2025, with Kalshi adding \$5.8 billion -- nearly \$10 billion combined monthly prediction market activity. Arbitrage bots alone earned an estimated \$40 million between April 2024 and April 2025. The market is large and growing, but also increasingly competitive.

Recommended Starting Point

For a new trading system: start with CEX-lag momentum on BTC/ETH 15-min Up/Down markets. The edge is documented, the implementation is straightforward (Binance WebSocket + py-clob-client), and the market is liquid. Layer in YES+NO arbitrage scanning as a secondary strategy to capture risk-free opportunities when they appear. Build toward market making once the execution infrastructure is mature.

4. Sources & References

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